CS102	**Spring 2023/24**	Project Group	**1C**
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Criteria	TA/Grader	Inst
Presentation		
Overall		

~ Combined ~

Pentagon

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Detailed Design

(Version 1.0)

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1. Introduction:

Combined is a multifunctional social media app Android application that is focused on fashion, targeting everyday usage and allows people to get in contact with each other. Sometimes it could be harder to understand the user interface for the users, so we tried to keep the level of the interface simple. Without GUI it is quite challenging so GUI takes a crucial part in our app. This app for now will focus on Bilkenters only. It can be difficult to manage and follow a social media app that doesn't focus on a specific community, which is why this app is exclusively for Bilkent students. Combined has two main features. Firstly, it allows you to share your combine on an online platform where others can view it. Secondly, it has ranking and comment pages that give others the chance to rate and comment on your combine out of 5. This is a great opportunity for you to get feedback and improve your combine.

2. System Overview

1. Frontend

1.1. Android Studio

As Combined will be an Android-based software, we are using Android Studio for developing the user interface and frontend. The layout editor feature significantly alleviates the burden on editors, and its integration with third-party software like Figma makes the design process much easier. Additionally, through the emulator, we can see how our application will appear on any device or how it will respond to specific inputs. Moreover, it can be easily integrated with Firebase, which we consider to be its biggest advantage for our database needs.

1.2. Firebase

Firebase is one of the most used mobile and web application development platforms among all the platforms Google offers. Since all Firebase services are integrated with others, it is one of the best options currently on the market. It provides useful tools to developers for application development, updating, auditing, viewing user performance, and much more!

1.2.1. Firebase Authentication

In social media applications, it is essential to know the identity of each user. Knowing the user's identity provides great flexibility for the user to access their account from different devices while reducing confusion while uploading each user’s data to the collective database. Firebase Authentication is an application that allows users to sign up and log in using their email and password. Additionally, after signing up, Firebase provides users with a unique user ID and a frequently changing token. Supporting popular unified identity providers like Google, Facebook, and Twitter makes it one of the best options. We will use Firebase Authentication on the sign-in page.

2. Data

2.1. Firebase

2.1.1. Cloud Storage

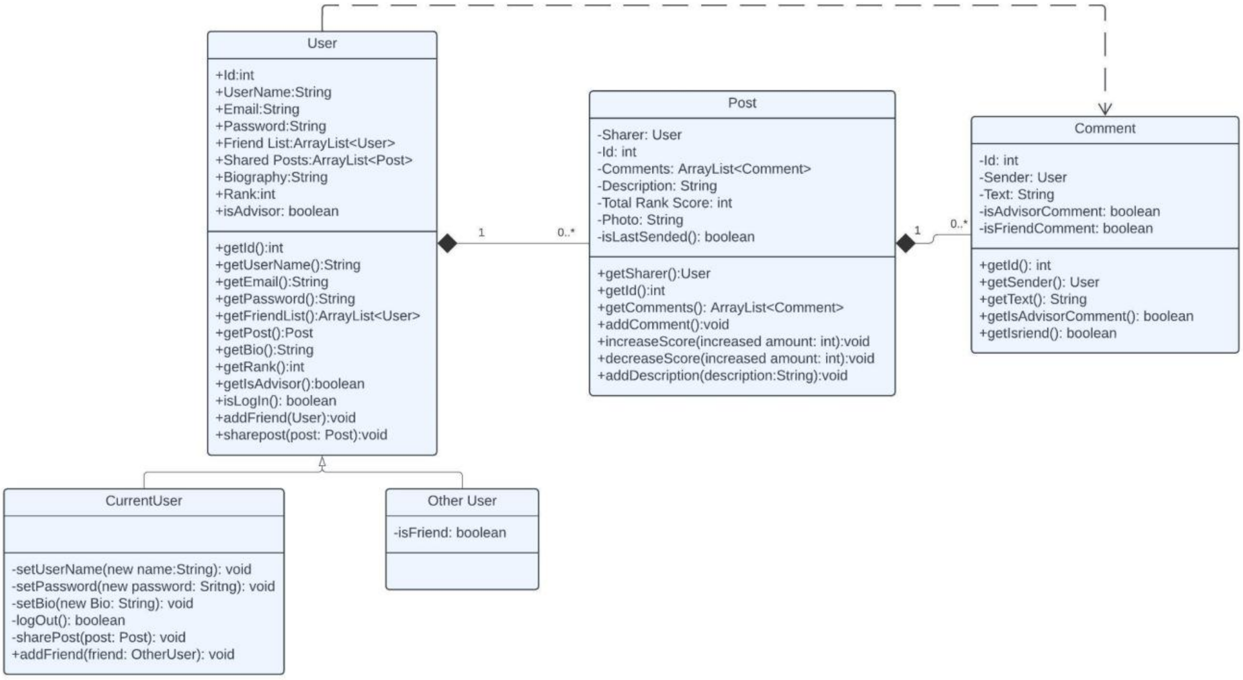
Firebase Cloud Storage is designed to store and serve user-generated content such as photos and videos. With its broad permission spectrum, it's easy to adjust who can access which file. Additionally, file upload and download processes occur independently of network quality, considering users without a stable internet connection. We will use Firebase Cloud Storage to securely store user posts and profile pictures.

2.1.2. Cloud Firestore

Being more flexible, scalable, serverless, and NoSQL are some of the features that distinguish Firestore from Cloud Storage. Data in Firestore is stored in collections, allowing developers to personalize them as they wish. Listeners in Firestore track changes happening in the application in real-time, and once the necessary operations are applied, the result is instantly reflected in the application interface. Similar to Cloud Storage, it also operates independently of network quality. We will use Cloud Firestore to store numerical and textual values in the comment, ranking, profile, and podium pages.

3. Design Details

3.1 Model Classes and Interfaces



The User class represents the user who is using this app, and it stores information about that user. Stored data are ID, username, email, password, biography, rank, and boolean for whether the user is an advisor or not. It also stores the user's friends list and old posts. The user class also has an inheritance relationship with classes CurrentUser and OtherUser. Those two subclasss differentiate who is the current user of that account. Additionally, this class has an aggregation relationship with the Post class.

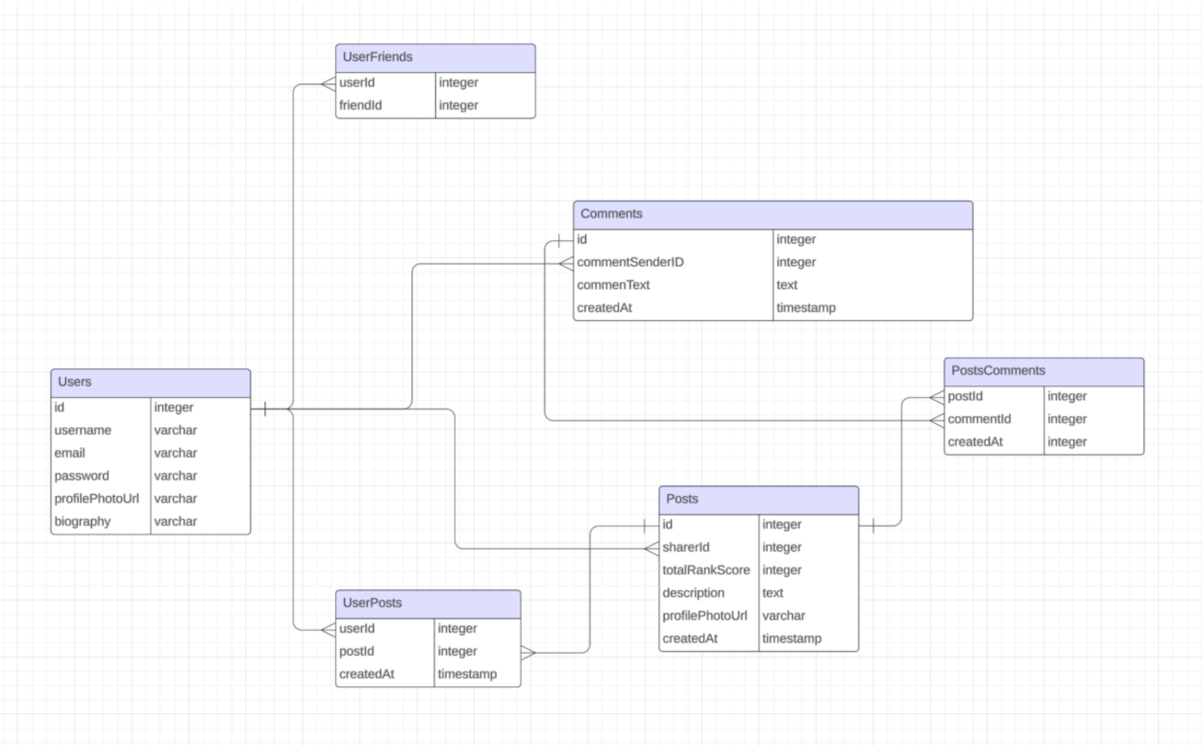
The first subclass is CurrentUser, which represents the account user. In these classes, users can make changes to the information, such as changing passwords or biographies and logging out from accounts. Also, the user can share posts and add friends.

The second subclass is OtherUser, which represents the users who have reached out from other accounts. The only information stored in this class is the user friend of the user who reached this OtherUser.

The Post class represents a post. It stores the information of post ID, sharer of the post, description, total score, photo, and boolean data, including whether it is the last sent post of the sharer. It also has a list of comments that hold the comment objects. This boolean is needed to determine whether the post is displayed on the other users' main page. Moreover, this class has an aggregation relationship with the comment class.

The comment class represents the comments of the post. Every comment class stores its own ID, sender, text, and boolean data to determine whether the comment sender is a friend of the displayer and whether the sender is an advisor. These boolean data are needed to organize the comment page. Since those boolean data depend on the user of the account, this comment class has a dependency relationship with the User class.

3.2 Database Schema



This SQL schema is designed to manage the functionalities of Combined. The schema comprises various tables, including Users, Posts, Comments, and associated features.

Users: This table stores various information about users, like their ID, username, email, password, profile photo URL, and personal biographies. It acts as the core focus of all user-related interactions inside the application.

Posts: This table contains details about the posts made by different users. Each post record has its own ID, sharer ID, total rank score, description, profile photo URL, and the timestamp of creation.

Comments: This table's main function is storing comments made on posts. It comprises an ID that is special for each comment, as well as the sender's ID, the text, and the timestamp.

UserFriends: An arrangement to monitor user friendships. It connects the friend's user ID to the user ID (both referencing Users).

UserPosts: With the help of the user ID, post ID (which refers to posts), and the creation timestamp, this table allows users to interact with their posts.

PostComments: This table links comments, post ID (referencing Posts), and comment ID (referencing Comments) to the posts to which they belong.

3.3 MVC Architecture

MVC is an acronym for Model-View-Controller, a software architecture pattern which makes it easier to manage an application by separating it to three concerns: model, view and controller. Model encapsulates the data and provides methods to manipulate them. Model responds to requests from the controller which are updating the data. We will specifically use "model" for the background information which users do not need to see. View presents the data to the user. It represents the UI elements such as text fields, images and buttons. We will use “view” structure for the visual parts of our application. Controller is the connection between model and view. Controller receives user input and implements actions accordingly. We will use “controller” structure for the user interactions with the application such as touching a button. In such a situation, the controller will handle the input, update the model and lastly, chooses the right view to be presented.

4. Task Assignment

UI Design	Şebnem Bihter Keleş, Nevzat Han Altın, Barış Peksak, Ahmet Ozan Annagür
Frontend	Şebnem Bihter Keleş, Ahmet Ozan Annagür
Database	Ege Şeşen, Barış Peksak

5. Summary & Conclusions

Combined is a social media platform designed specifically for Bilkent students. It has a versatile and user friendly interface with a focus on simplicity and user convenience. The app is built using Android Studio and Firebase mainly which provides a strong foundation for both the front-end and back-end of the app.

Combined integrates Firebase Authentication, Cloud Storage, and Cloud Firestore to guarantee security, scalability, and real-time data synchronization. This ensures a smooth user experience, even under poor network conditions. The app is designed to manage data effectively and foster meaningful interactions among users.

The app's classes and database are meticulously designed to ensure efficient data management and interaction, featuring well-defined entities such as User, Post, and Comment that encapsulate essential user information.

Briefly; Combined is an innovative and collaborative platform that specifically allows Bilkent students to keep in touch with each other.